

Objective Hardness Geometry in Deepfake Speech Detectors

Paper Plan & Evidence Map

A planning document for deciding what the paper is — not a draft. Primary narrative: Story 3.
Figures generated from repository results are embedded throughout.

Primary contribution:	Story 3 — geometric hardness factor C (causal, actionable)
Factor weighting:	C : T = 70 : 30 (T real but fragile; cautious)
ASVspoof role:	Causal validation via E9 only
Excluded:	E8 (confounded ASVspoof correlation)
Story 1:	Supporting mechanism + supplement
Venue framing:	Dual: Speech (Interspeech/ICASSP) + ML-interp (NeurIPS/ICLR)

Generated 2026-06-09 · companion to [outputs/paper_plan_story3.html](#) · supporting assets organised under [final_outputs/{mandatory,optional}/](#)

1 · Executive Summary

Some text-to-speech / voice-conversion systems are systematically hard for detectors — and that hardness is not random. We argue it has an objective geometric origin: the compactness of a system's speech trajectory in deep self-supervised (WavLM) space. We name this factor C and show it (i) predicts per-system hardness, (ii) is causal under direct intervention, and (iii) is actionable — targeting it lowers EER on the hardest systems.

Factor definitions

- C (compactness) = $-\text{rog@L12}$ — negative radius of gyration of the WavLM layer-12 frame cloud. Higher C = more compact. Primary factor.
- T (trajectory irregularity) = vel_entropy@L9 — entropy of frame-to-frame velocity at layer 9. Secondary (70/30), fragile.

What's strong (lead with these)

- Hardness is system-intrinsic — systems agree on what is hard across seeds ($p \approx 0.79\text{--}0.84$).
- $C \leftrightarrow$ hardness on MLAAD (63 systems): $\rho(C, \text{resid}) = +0.29$; more compact $\rightarrow$ harder.
- C is causal (E9): scaling compactness moves EER through baseline, beating a matched-energy shift control.
- C transfers to in-the-wild at the speaker level: $\rho(C, \text{EER}) = +0.44$ (45 speakers).
- C is actionable: hardness reweighting (P5) lowers the hardest (C-Q4) systems' EER by 0.023 (AUC +0.026).

What's fragile (handle carefully)

- T is fragile — scale-dependent, not independently manipulable in E9, likely distorted in in-the-wild audio.
- Encoder scope — C validates on WavLM & wav2vec2 but FAILS on HuBERT. Never claim "all SSL."
- Coverage — C+T explain $\approx 9\%$ of residual-hardness variance (LOO). Frame as "actionable factors," not "we explain hardness."

Verdict on Story 1 (GAT-hierarchy interpretability): mechanistically rich but narrow (ASVspoof-only, language-invariance untested). Belongs as a compressed supporting mechanism + supplement — not a pillar.

2 · Narrative Recommendation

Keep Story 3 as the spine. The call on every candidate sub-story:

Sub-story	Placement	Support	Completes vs distracts
Objective hardness agreement (seeds)	MAIN	STRONG	Completes — establishes the phenomenon is real
C & T as geometric factors (MLAAD)	MAIN	STRONG (C)	Completes — the core contribution
C causality via intervention (E9)	MAIN	STRONG	Completes — keystone above correlation
Actionability (P1 / P5)	MAIN	STRONG	Completes — the 'so what'
In-the-wild support for C	MAIN	SUPPORTING	Completes — external validity
Temporal invariance / smoothing	APPENDIX (+1 fig)	MEDIUM	Completes — motivates T; hardness targetable
Story 1 — GAT hierarchy	APPENDIX (+1 fig)	MEDIUM	Completes — a 'why' mechanism, if compressed
Story 2 — language overfitting	OMIT	WEAK	Distracts — one language pair; other paper
Story 4 — taxonomy fragility	OMIT	WEAK	Distracts — negative replication; other paper

Reframe required: the smoothing sub-story

The earlier framing ("smoothing improved detection") is wrong — but it is also not a failure. Smoothing-as-augmentation (P4) is a dose-dependent per-system lever: applied uniformly the mean regresses (EER $\Delta = +0.037$), yet 8 systems improve on both EER and AUC (MegaTTS3 -0.134 , Resemble.ai -0.078 , FireRedTTS-2.0 -0.072). It overcorrects systems that don't need it and helps those that do $\Rightarrow$ hardness is targetable. Separately, smoothing-as-intervention establishes causality (MA $k=3 \rightarrow k=10$: EER $0.421 \rightarrow 0.571$). Do not write "smoothing uniformly improved detection," and do not write "smoothing augmentation failed."

3 · Story 3 as the Primary Framing

Thesis. Deepfake-detection hardness has an objective, measurable geometric origin in deep SSL space: systems whose layer-12 WavLM trajectories are more compact (C) are systematically harder to detect; this factor is causal and can be acted on. A secondary dynamic factor (T) correlates with hardness but is fragile.

Evidence arc (paper spine)

- Objective — hardness rankings agree across seeds → not model noise.
- Geometric correlate — C predicts per-system hardness on MLAAD; T adds a weaker, correlated signal.
- Independent — $C \perp T$ ($r \approx +0.11$; separate PCA components; positive unique partials).
- Causal — intervening on C inside the frozen detector moves EER beyond a matched-energy shift control (E9).
- Actionable — C/T-guided calibration (P1) and reweighting (P5) lower hardest-system EER.
- External validity — C transfers to in-the-wild at the speaker level; T treated cautiously there.

Dual-venue note — Speech: foreground EER tables, benchmarks, actionability. ML/interp: foreground the causal-intervention design (E9) and representation-geometry framing.

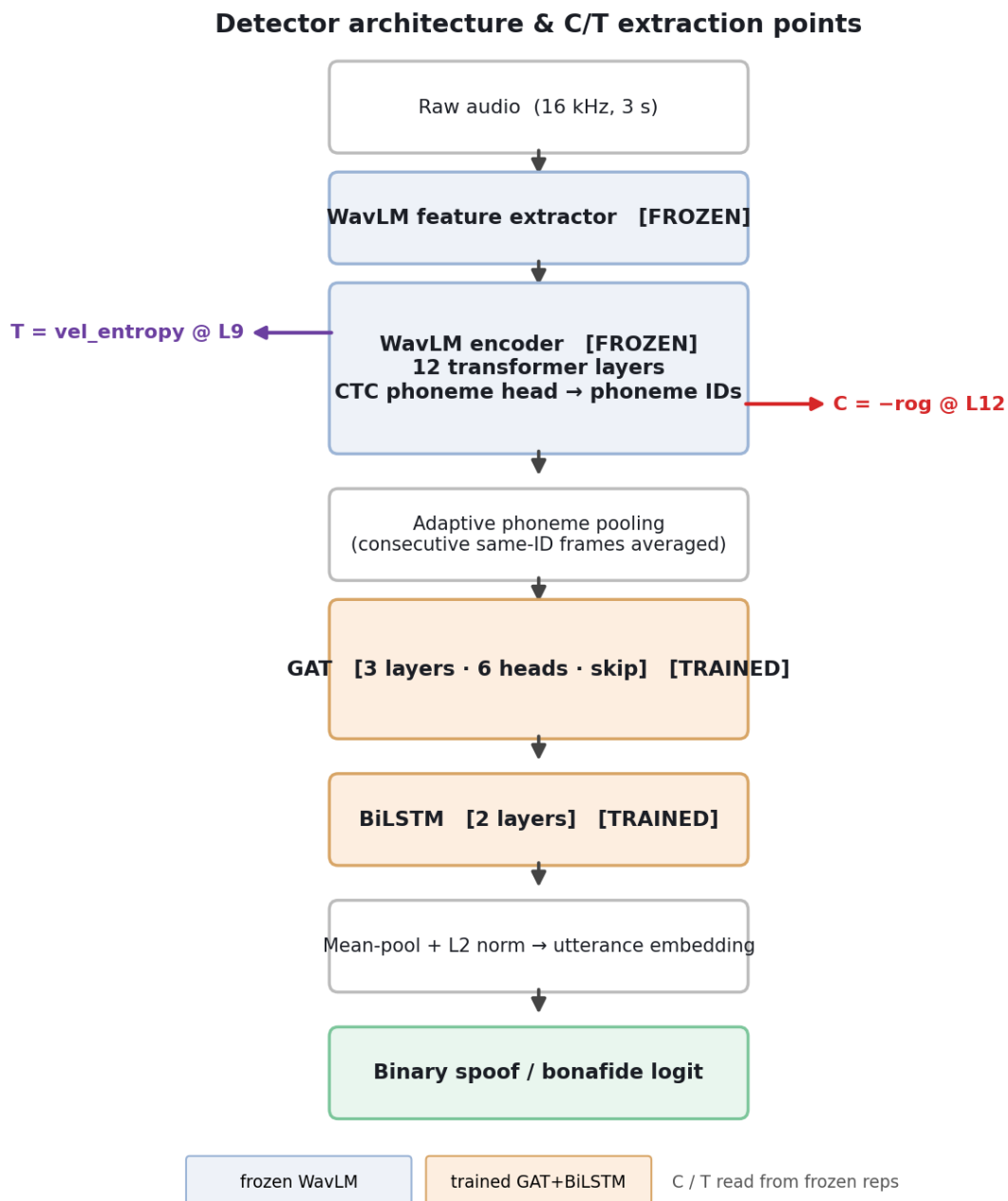


Figure 1. Detector architecture and the C/T extraction points: a frozen WavLM encoder feeds a trained phoneme-GAT + BiLSTM; C is read as $-\text{rog}$ at layer 12, T as velocity-entropy at layer 9.

4 · Story 1 Honest Review

Core contribution if primary: a hierarchical functional dissociation in a phoneme-GAT — early layers linearly encode vocoder/system identity (multiclass probe 91.8%_{@gat_l0} → 75%_{@bilstm}) while binary detection strengthens with depth; activation patching shows vowels/diphthongs drive the binary decision ($\Delta \approx +1.95$) whereas sibilants/nasals drive system-ID ($\Delta \approx +0.004$).

Question	Answer
Strong enough alone?	No — ASVspoof A01–A06 only (6 systems); tiny causal shifts (± 0.004); language-invariance untested.
Best role	Mechanistic explanation supporting Story 3 (why compact systems resist separation) + supplement.
Genuinely novel	Fingerprint-destruction $\leftrightarrow$ detection-creation layer trade-off; dual-subspace phoneme-class causal dissociation.
Speculative now	Language-invariance; cross-dataset generality; that destruction is 'purposeful'.
Experiments to add	Probe hierarchy + activation patching on MLAAD by language; intermediate-layer (L1) patching.
Into main paper	≤ 1 composite figure (probe hierarchy + phoneme-class causal bars).

Final verdict: KEEP AS SUPPORTING EVIDENCE (compressed mechanism section for Story 3); bulk to supplement.

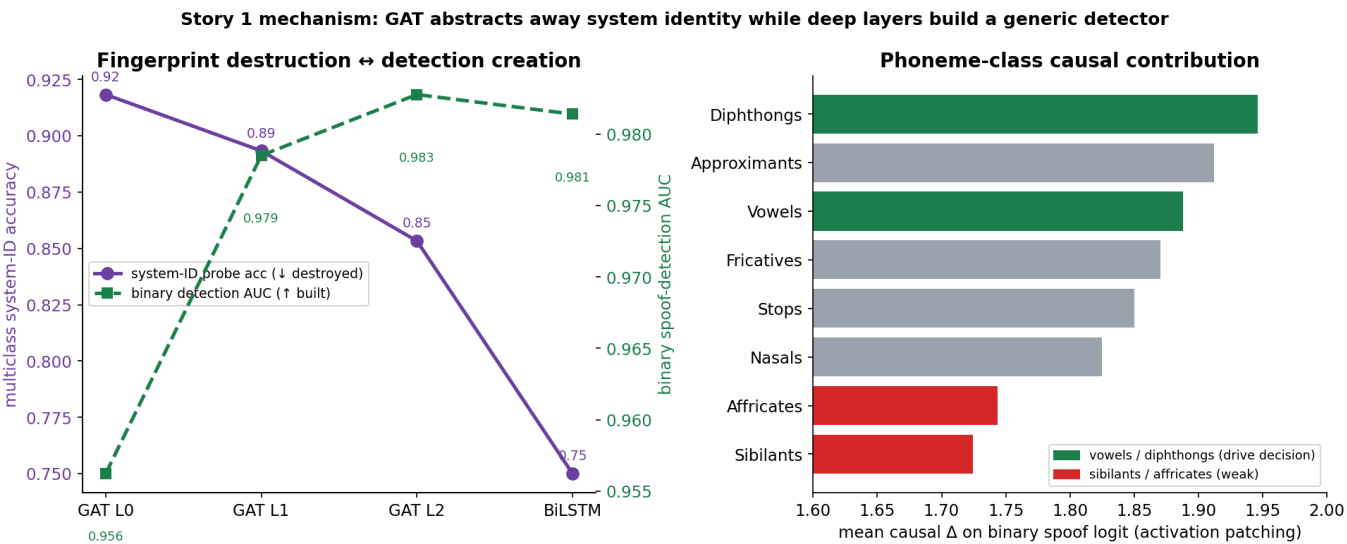


Figure 2. Story 1 mechanism. Left: system-ID probe accuracy falls L0→BiLSTM while binary detection AUC rises (fingerprint destruction $\leftrightarrow$ detection creation). Right: phoneme-class causal contribution — vowels/diphthongs drive the binary decision; sibilants are causally weak.

5 · Core Evidence Figures (main paper)

Deepfake hardness is objective: systems agree on what is hard across random seeds

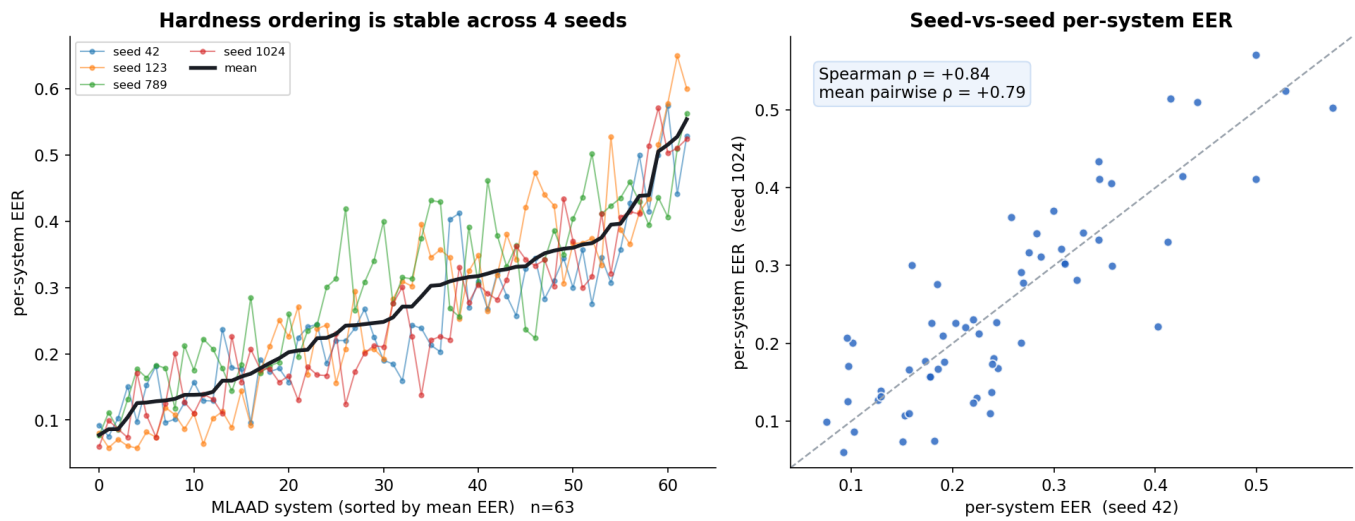


Figure 3. Hardness is objective. Per-system EER ordering is stable across random seeds (left); seed-vs-seed Spearman $\rho = +0.84$, mean pairwise $\rho = +0.79$ (right).

More compact (higher C) → harder, consistently across corpora

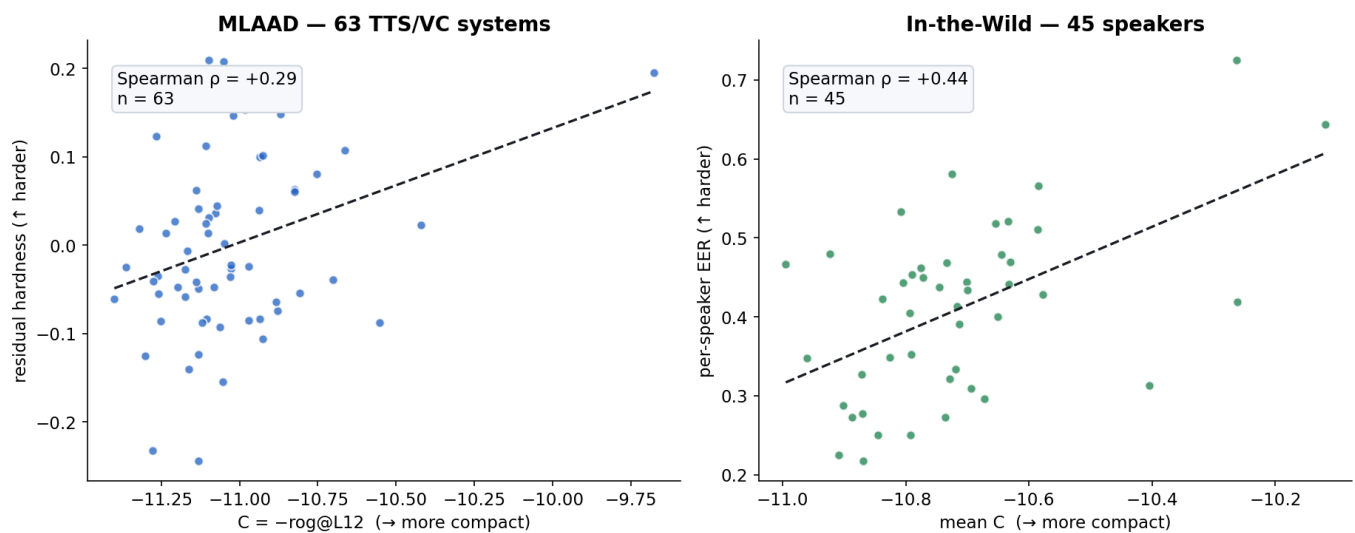


Figure 4. More compact (higher C) → harder, consistently across corpora: MLAAD systems $\rho = +0.29$ and in-the-wild speakers $\rho = +0.44$. (ASVspoof's opposite correlation, E8, is excluded.)

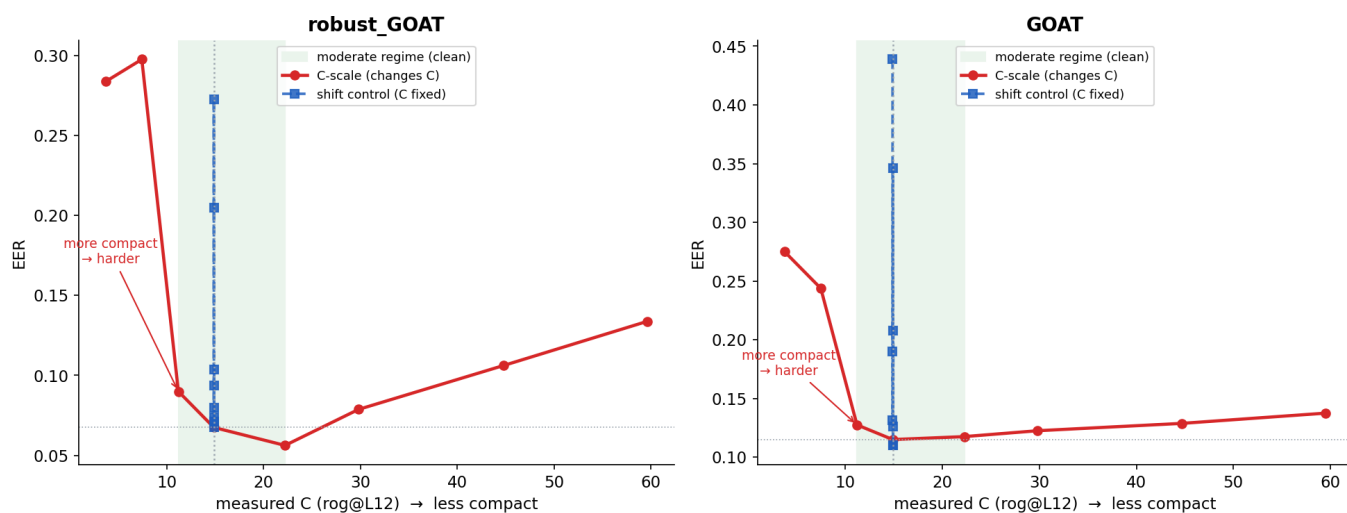
C is causal: scaling compactness moves EER beyond a matched-energy shift control

Figure 5. *C* is causal. Scaling compactness inside the frozen detector (red) moves EER through baseline and beats the matched-energy shift control (blue) in the clean moderate regime; both detectors agree. More compact → harder.

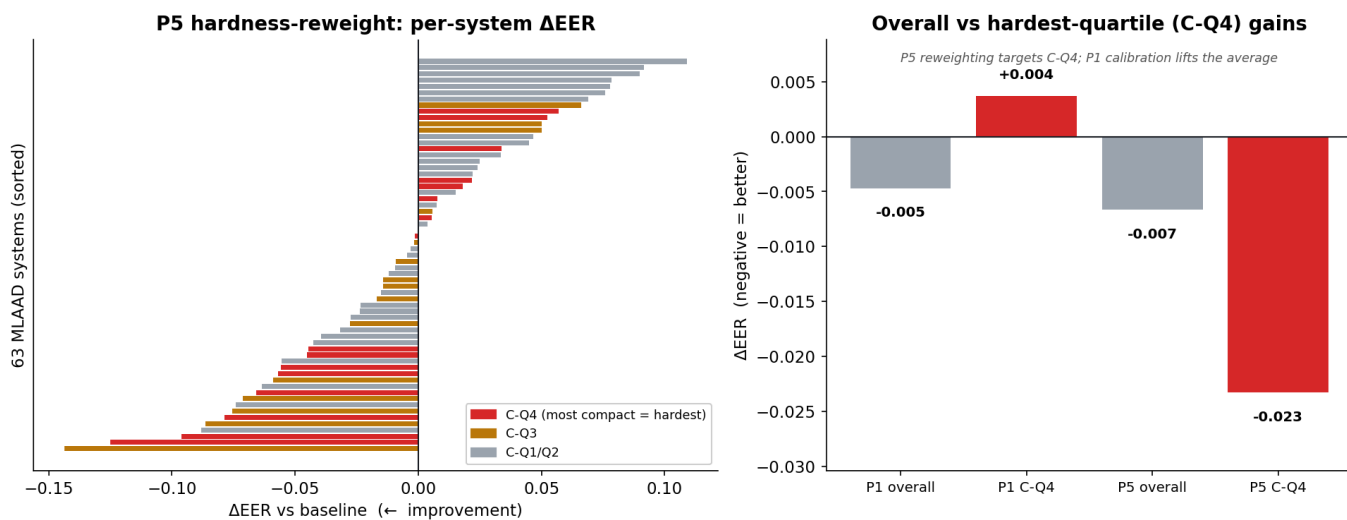
Hardness reweighting (P5) lowers EER on the hardest (C-Q4) systems; calibration (P1) improves the average

Figure 6. Actionability. Hardness reweighting (P5) lowers EER on the hardest C-Q4 systems (-0.023); calibration (P1) improves the average but not specifically C-Q4 (shown honestly).

**Uniform smoothing augmentation is a dose-dependent lever:
it genuinely helps the subset that needs it, and overcorrects the rest**

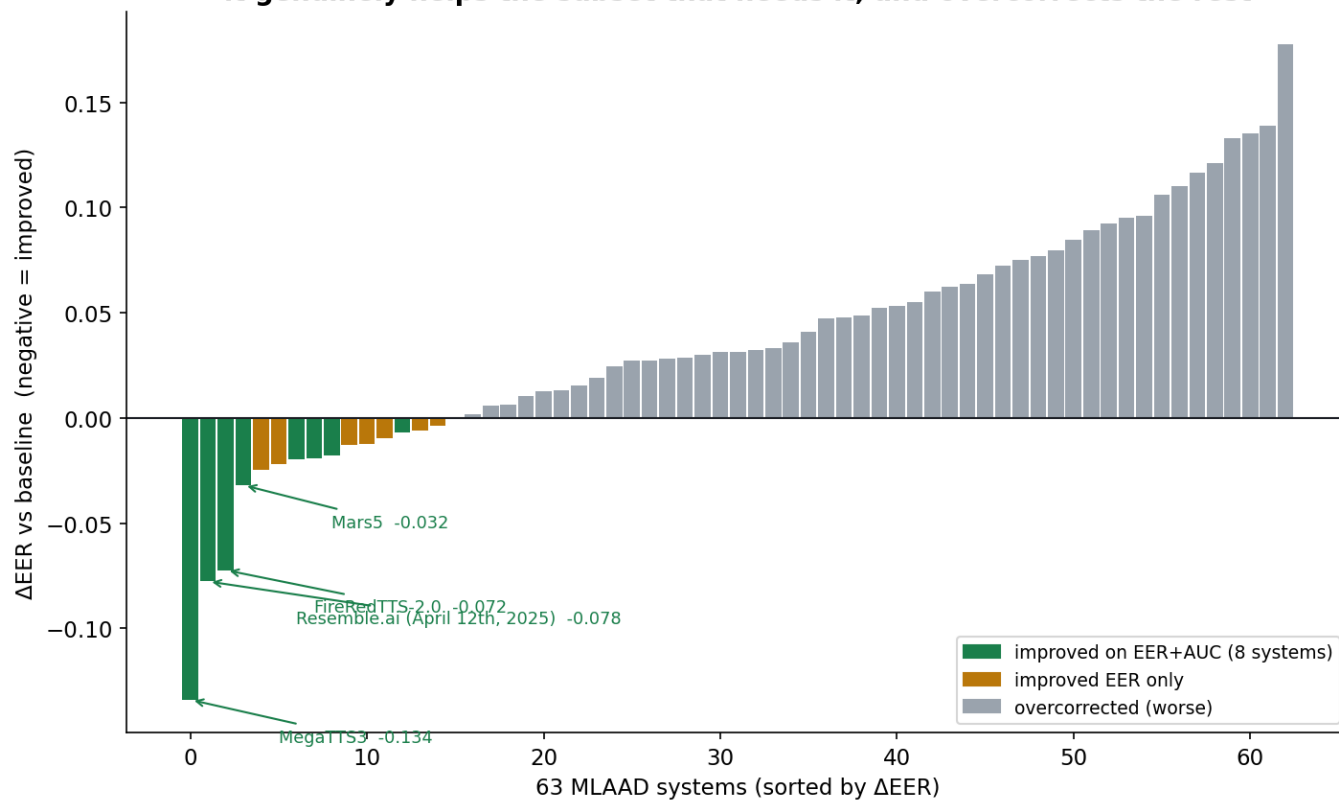


Figure 7. Smoothing augmentation is a dose-dependent lever: 8 systems genuinely improve on EER+AUC (MegaTTS3 -0.134) while the rest are overcorrected — evidence that hardness is targetable. (Appendix.)

6 · Claims Strength

Core claims — STRONG evidence

Claim	Basis
Hardness is system-intrinsic across seeds	seed ranking agreement, $p \approx 0.79\text{--}0.84$ (5 seeds)
C correlates with per-system hardness (MLAAD)	$p(C, \text{resid}) = +0.29$; $r \approx +0.35$, LOO $R^2 \approx 0.069$ (63 systems)
C is causal within the detector	E9: moderate regime EER $0.067 \rightarrow 0.090$ (compact) / $\rightarrow 0.056$ (expanded), beats shift control
C is actionable on hardest systems	P5 C-Q4 EER -0.023 , AUC $+0.026$; T-Q4 EER -0.020 , AUC $+0.027$
C and T are independent factors	$r(C, T) \approx +0.11$; separate PCA components; positive unique partials

Supporting claims

Claim	Basis
C transfers to in-the-wild (speaker level)	$p(C, \text{EER}) = +0.44$, $n=45$; mostly between-speaker
Temporal smoothness is a hardness factor	smoothing intervention MA $k=3 \rightarrow k=10$: EER $0.421 \rightarrow 0.571$
Smoothing is a dose-dependent lever	P4: 8 systems improve on EER+AUC (MegaTTS3 -0.134)
C generalizes across encoders (partial)	WavLM $r = +0.348$, wav2vec2 $r = +0.346$
Adversarial training exploits C	robustness-mechanism analysis (ΔEER vs C; head entropy collapse)

Claims to avoid overstating

Tempting claim	Say instead
T is a robust universal factor	T is correlated but fragile; near-constant in ASVspoof $\rightarrow$ neither proven nor refuted there
C generalizes across all SSL encoders	C validates on WavLM & wav2vec2; fails on HuBERT
C predicts hardness on ASVspoof (corr.)	Omit — ASVspoof = causal E9 only; E8 excluded
Smoothing improves detection	Removing temporal variance hurts; augmentation is a targeted lever
The GAT hierarchy is universal	Demonstrated on ASVspoof A01–A06; generality untested
We explain deepfake hardness	We identify actionable factors; $\approx 65\%$ residual unexplained

Six reviewer danger zones — keep claims tight: cross-dataset · cross-encoder (HuBERT) · in-the-wild (codec $\rightarrow$ T) · temporal smoothing · GAT-mechanism generality · explained variance.

7 · Supporting Assets — final_outputs/

Every figure, result and summary is grouped by sub-narrative under final_outputs/. MANDATORY = main-paper evidence; OPTIONAL = supplement.

mandatory/ (7 sub-narratives)

Folder	Key contents
00_architecture	fig1 schematic (svg+png), research.md
01_objective_hardness_seed_ranking	fig_hardness_seed_agreement, seed_stability_summary.json, per-seed EER csv
02_geometry_C_and_T	C→hardness, orthogonality, layerwise-R ² , E1 cross-encoder; final_report.md, orthog report
03_causality_E9	fig_e9_causal_C_main, e9_summary.md, e9_causal_C.csv
04_actionability_P1_P5	fig_actionability, P1/P5 summaries + per-system csvs
05_in_the_wild_C	e6_itw_speaker_ct_eer, E5/E6 ITW summaries + speaker CT csv
06_mechanism_story1_composite	fig_story1_mechanism + probe/patching stats csvs

optional/ (9 sub-narratives)

Folder	Key contents
temporal_smoothing	dose-dependent + intervention figs; p4 summary; causal_interpretation.md
geometry_supplement	PCA / variance-partition / Shapley / interaction / E2 / E3 + summaries
causality_replication_and_T	E9 GOAT replication; E9 T-intervention figs + csv
in_the_wild_supplement	ITW manifold / distributions / within-between / quartiles
actionability_supplement	P1 extra panels; P3/P6/P7/P8 summaries
robustness_mechanism_E7	ΔEER-vs-C, quartile-C, AUC-vs-C + robustness summary
manifold_hardness_landscape	manifold geometry, extreme-system, residual-hardness panels
story1_full_mechanism	probe rankings, activation patching, attention-KL, PC1, etc.
e4_asvspoof_inconclusive	E4 A01-A06 figure + summary (low usefulness)

8 · Related Work — Mandatory Citations

26 works a credible related-work section must engage, grouped by theme. Bibliographic details verified against arXiv / proceedings (June 2026).

Self-supervised speech encoders

- [1] Baevski et al., 2020. wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations. NeurIPS.
- [2] Hsu et al., 2021. HuBERT: Self-Supervised Speech Representation Learning by Masked Prediction of Hidden Units. IEEE/ACM TASLP.
- [3] Chen et al., 2022. WavLM: Large-Scale Self-Supervised Pre-Training for Full Stack Speech Processing. IEEE JSTSP.
- [4] Babu et al., 2022. XLS-R: Self-Supervised Cross-Lingual Speech Representation Learning at Scale. Interspeech.

SSL probing & layer-wise analysis

- [5] Yang et al., 2021. SUPERB: Speech Processing Universal PERFORMANCE Benchmark. Interspeech.
- [6] Pasad, Chou & Livescu, 2021. Layer-wise Analysis of a Self-Supervised Speech Representation Model. IEEE ASRU.
- [7] Pasad, Shi & Livescu, 2023. Comparative Layer-wise Analysis of Self-Supervised Speech Models. ICASSP.
- [8] Alain & Bengio, 2017. Understanding Intermediate Layers Using Linear Classifier Probes. ICLR Workshop.
- [9] Hewitt & Liang, 2019. Designing and Interpreting Probes with Control Tasks. EMNLP.

Anti-spoofing corpora & detectors

- [10] Wang et al., 2020. ASVspoof 2019: A Large-Scale Public Database of Synthesized, Converted and Replayed Speech. Computer Speech & Language.
- [11] Yamagishi et al., 2021. ASVspoof 2021: Accelerating Progress in Spoofed and Deepfake Speech Detection. ASVspoof Workshop.
- [12] Müller et al., 2022. Does Audio Deepfake Detection Generalize? (In-the-Wild dataset). Interspeech. arXiv:2203.16263.
- [13] Müller et al., 2024. MLAAD: The Multi-Language Audio Anti-Spoofing Dataset. IJCNN. arXiv:2401.09512.
- [14] Jung et al., 2022. AASIST: Audio Anti-Spoofing Using Integrated Spectro-Temporal Graph Attention Networks. ICASSP.
- [15] Tak et al., 2021. End-to-End Anti-Spoofing with RawNet2. ICASSP.
- [16] Tak et al., 2022. ASV Spoofing and Deepfake Detection Using wav2vec 2.0 and Data Augmentation. Odyssey. arXiv:2202.12233.
- [17] Tak et al., 2022. RawBoost: A Raw Data Boosting and Augmentation Method Applied to ASV Anti-Spoofing. ICASSP.

Graph attention & mechanistic interpretability

- [18] Veličković et al., 2018. Graph Attention Networks. ICLR.
- [19] Vig et al., 2020. Investigating Gender Bias in Language Models Using Causal Mediation Analysis. NeurIPS. arXiv:2004.12265.
- [20] Meng et al., 2022. Locating and Editing Factual Associations in GPT (ROME / activation patching). NeurIPS. arXiv:2202.05262.

Representation geometry & instance hardness

- [21] Pappas, Han & Donoho, 2020. Prevalence of Neural Collapse during the Terminal Phase of Deep Learning Training. PNAS. arXiv:2008.08186.
- [22] Ansuini et al., 2019. Intrinsic Dimension of Data Representations in Deep Neural Networks. NeurIPS. arXiv:1905.12784.
- [23] Smith, Martinez & Giraud-Carrier, 2014. An Instance Level Analysis of Data Complexity (instance hardness). Machine Learning.
- [24] Bengio et al., 2009. Curriculum Learning. ICML.
- [25] Shrivastava, Gupta & Girshick, 2016. Training Region-based Object Detectors with Online Hard Example Mining (OHEM). CVPR.
- [26] Swayamdipta et al., 2020. Dataset Cartography: Mapping and Diagnosing Datasets with Training Dynamics. EMNLP.

Venue tilt — Speech: foreground refs 10–17 (ASVspoof/AASIST/RawBoost). ML/interp: foreground refs 6–9, 18–22.

9 · Final Recommendation

Write one paper, Story 3, organized around C — a geometric hardness factor that is objective (seed-agreement), causal (E9 beats a matched-energy control) and actionable (P1/P5) — with T as a 70/30 fragile companion we caveat but do not denounce.

- Corpus roles: MLAAD = correlational + actionability backbone; ASVspoof = causal keystone via E9 only (E8 excluded); in-the-wild = external validity for C.
- Story 1 → one compressed mechanism section + supplement (supporting, not a pillar).
- Drop Stories 2 & 4 to a future paper.
- Guard the six danger zones; the seven mandatory-folder sub-narratives are the main-paper backbone.